Beyond Correctness: Causally Grounded Generative Student Modeling with Recurrent and Intervenable Student States

Anonymous authors
Paper under double-blind review

Abstract

Student modeling is commonly divided into separate problems: predicting correctness, forecasting a student's concrete response, diagnosing a misconception, and estimating future behavior. This fragmentation makes it difficult to reuse evidence across tasks and limits most models to a fixed label space or a task-specific prediction head. We introduce CGSM, a causally grounded generative student model that expresses these problems through one conditional-generation interface and one recurrent student state. After every observed interaction, a language-model updater compresses the previous state and the new evidence into a continuous state carried by a small bank of eight state slots addressed by one special token type. A task-conditioned decoder predicts correctness, responses, misconceptions, prospective responses, or error recurrence from this state and the query, without direct access to the history. To make the state predictive rather than merely decodable, we combine multi-task generation with full-context distillation, recurrent–direct state consistency, counterfactual supervision from executable misconceptions, and a future-profile ownership objective. We evaluate the model across six educational data sources and separately test predictive sufficiency, temporal updating, readable future profiles, state-ownership permutations, and within-student stability. The resulting framework turns student state from a post-hoc activation into a compact computational object that can be cached, updated, read, and audited.

1 Introduction

What does a student know, and what will they do next? Educational systems usually operationalize this question as a collection of separate prediction problems. Knowledge tracing estimates whether the next response will be correct (Corbett & Anderson, 1995; Piech et al., 2015; Ghosh et al., 2020); cognitive diagnosis associates performance with latent skills; response models predict a selected option; and misconception systems explain a particular error. These targets are related, but the models used to predict them rarely share an interface. A system that outputs a scalar probability of correctness cannot directly say that a student is likely to add two denominators, produce the resulting fraction, or preserve that error on a later problem without adding a new task-specific output space.

This separation matters because correctness is a lossy description of student behavior. Two students can both answer incorrectly while producing different responses for different reasons, and those differences determine which future errors are plausible and which feedback would be appropriate. Recent datasets make richer modeling possible: Eedi links distractors to expert-authored misconceptions (Wang et al., 2021); MalruleLib provides executable mathematical error rules (Chen et al., 2026); and FoundationalASSIST contains longitudinal problem text together with students' first responses (Worden et al., 2026). Yet the dominant knowledge-tracing formulation still maps an interaction sequence to a fixed correctness head.

Large language models (LLMs) offer a natural way to express heterogeneous targets as conditional generation. They can consume problem text, answer choices, skill descriptions, and student histories in a shared context, then emit either a class label or an open-form response. However, using an LLM does not by itself produce a valid student model. A capable language model may solve the problem rather than imitate the student, rely on item priors rather than the student's history, or generate a plausible explanation unrelated to the student's observed error. Prior work has accordingly questioned whether LLMs can simulate students under perturbed evidence (Sonkar & Baraniuk, 2023) and whether apparent student simulations remain valid under stronger controls (Yuan et al., 2026). A useful generative student model must therefore pass both predictive and mechanistic tests: it should use student-specific evidence to predict multiple outcomes, and its internal student representation should have a demonstrable effect on those predictions.

We introduce CGSM, a causally grounded conditional-generation model with an explicit recurrent student state. After each observed interaction, a shared language-model updater transforms the previous continuous state and the new evidence into z_t. A task-conditioned decoder then receives only z_t and a query; the raw history is unavailable as a bypass. The same model produces structured outputs for five constructs: (i) next-response correctness, (ii) the student's concrete response, (iii) the misconception expressed by an observed error, (iv) correctness and response on a future problem after unobserved intervening interactions, and (v) the recurrence of the same wrong response on a later exposure. The formulation includes both closed-choice and open-generation protocols; it does not average their unlike metrics into a single "student modeling score."

The main methodological challenge is to make z_t sufficient for prediction without sacrificing the performance of a full-context language model. We use four complementary objectives. First, a full-context teacher distills its output distribution into the state-only decoder. Second, a trajectory consistency loss aligns recurrently updated states with states independently encoded from the corresponding complete history. Third, executable misconceptions provide counterfactual supervision: swapping a donor student's state into a recipient query should generate the error obtained by applying the donor's rule. The last objective grounds state interventions behaviorally rather than training an auxiliary misconception classifier. Fourth, a future-profile objective aligns a frozen view of the state with later knowledge-component outcomes while teaching matched student–state ownership.

We evaluate the resulting state with a student-disjoint longitudinal protocol. A readout selected only on training and validation students predicts held-out future knowledge-component rates and emits readable weakness profiles. We compare it with a population prior and a same-window correctness/count baseline, then permute state ownership across test students and compare two disjoint views of each student's observed history. The permutation asks whether the right student's state matters; the disjoint-view test asks whether the recovered information is stable. Literal cross-student text transport is reported as a diagnostic rather than a primary gate because an arbitrary transplanted latent trajectory need not be in distribution. This distinction is essential because decodability alone does not establish a causal role (Hewitt & Liang, 2019; Elazar et al., 2021).

We instantiate the framework with a 32B-parameter language model and a mixture of six data sources spanning real longitudinal interaction logs, expert-labeled misconceptions, and executable synthetic error rules. Our formal evaluation keeps students disjoint across longitudinal splits and separates temporal prediction, non-temporal profile prediction, full-taxonomy misconception retrieval, open generation, and controlled synthetic generalization. This protocol is deliberately stricter than evaluating against a row-specific set of candidates known to contain the gold answer. It also makes the limits of each construct explicit: future performance within a knowledge component is not treated as a direct mastery label, and repeated incorrectness without a recorded response is not treated as persistence of the same misconception.

Our contributions are:

- A recurrent generative student state. We cast correctness, concrete response prediction, misconception diagnosis, prospective behavior, and exact wrong-response recurrence as task-conditioned generation from one continuously updated state.
- A construct-aware multi-source evaluation. We align six educational data sources while preserving source-specific semantics and compare the unified model with classical and neural specialists, including IRT, PFA, BKT, DKT, AKT, retrieval, and response-frequency models.
- Transfer across synthetic and authentic errors. We test whether supervision from executable misconception rules improves diagnosis of expert-labeled errors, with task-matched ablations that distinguish rule supervision from merely adding synthetic responses.
- Grounded state training. Full-context distillation preserves predictive information, trajectory consistency teaches incremental updating, executable counterfactuals supervise task behavior, and future profile supervision binds states to student-specific outcomes.
- A reusable, audited student state. We combine held-out future prediction, readable profiles, state-ownership permutations, matched baselines, and disjoint-history stability to test whether the learned state is useful and student-specific.

Together, these contributions shift the question from whether a language model can predict one educational label to whether one model can represent a student's behavior at several levels of specificity—and whether the internal state supporting those predictions can survive causal scrutiny.

2 Related Work

2.1 Knowledge tracing and student performance prediction

Knowledge tracing (KT) estimates a student's evolving knowledge from a sequence of learning interactions. Bayesian Knowledge Tracing represents mastery of a knowledge component with a hidden Markov model (Corbett & Anderson, 1995), while Performance Factors Analysis uses counts of prior successes and failures as interpretable predictors (Pavlik et al., 2009). Item-response and factor models instead explain outcomes through student ability and item difficulty (Rasch, 1960; Vie & Kashima, 2019). Neural KT replaces hand-specified transitions with recurrent networks (Piech et al., 2015), external memories (Zhang et al., 2017), or self-attention (Pandey & Karypis, 2019; Ghosh et al., 2020). These models are strong and appropriate baselines for correctness prediction, but their standard objective and output space remain task-specific: they estimate a binary outcome for an indexed item or knowledge component. Our work does not argue that such models are incapable of richer prediction after modification. Instead, we ask whether correctness, concrete responses, misconception text, and prospective responses can be handled by one generative interface without adding a separate prediction head for every target.

The distinction also motivates our evaluation design. Standard KT benchmarks often emphasize warm-item, new-student prediction, where test students are held out but most test items occurred during training. Text-rich datasets such as XES3G5M (Liu et al., 2023), DBE-KT22 (Abdelrahman et al., 2022), and FoundationalASSIST (Worden et al., 2026) permit models to use problem content in addition to identifiers. We retain student-disjoint temporal tests but report protocols by data source and prediction type rather than pooling unlike label spaces into a macro-average.

2.2 Responses, misconceptions, and cognitive diagnosis

Correctness does not identify the process that generated an answer. Cognitive diagnosis models infer mastery profiles over a predefined skill matrix (de la Torre, 2009), while distractor and misconception modeling links particular wrong answers to systematic gaps in understanding. The Diagnostic Questions/Eedi benchmark made this connection explicit

by associating mathematics distractors with expert-authored misconceptions (Wang et al., 2021). More recently, MalruleLib formalized 101 executable mathematical error rules and their step traces (Chen et al., 2026). These resources support complementary forms of evidence: Eedi provides authentic expert labels for observed distractors, whereas MalruleLib provides controlled counterfactual behavior and exact rule identity.

Our misconception task differs from ordinary per-example candidate ranking. The main protocol either retrieves from the complete misconception taxonomy or generates a diagnosis without exposing a candidate set constructed to contain the gold label. Our response tasks similarly distinguish closed-choice prediction from open mathematical response generation. This separation is important: retrieving a supplied misconception and generating a student's future response are not interchangeable capabilities, even when both can be scored with semantic similarity.

2.3 Language models as knowledge tracers and simulated students

Recent work incorporates pretrained language models into KT to represent question semantics and address cold-start items. Language Model Can Do Knowledge Tracing adapts language-model representations to KT (Lee et al., 2024); CLST aligns a generative language model with a student tracer for cold-start mitigation (Jung et al., 2024); and SINKT combines language-model semantics with structural information for inductive KT (Fu et al., 2024). This line of work primarily retains correctness as the prediction target. It demonstrates the value of language semantics for KT but does not by itself unify correctness with open responses and misconception diagnosis.

A related line asks LLMs to act as students. Sonkar and Baraniuk (2023) probe whether LLMs follow provided student evidence under perturbations. Agent-based approaches condition LLMs on coarse cognitive levels to generate imperfect behavior (Wu et al., 2025). Contemporaneous work emphasizes that plausible generations are insufficient evidence of valid student simulation and calls for comparisons against real student behavior and stronger controls (Yuan et al., 2026). We share this concern but study a different setting: the model is trained over actual student histories and multiple predictive targets, rather than prompted to enact a persona or ability level. We also test whether history-dependent information is localized in an internal state that can be intervened on.

2.4 Probing and intervening on internal representations

Linear probes are widely used to identify information encoded in neural representations, but probe accuracy can reflect information that the model does not use for its prediction. Control tasks and selectivity analyses help separate representation content from probe capacity (Hewitt & Liang, 2019), while counterfactual removal or intervention tests whether an encoded feature is behaviorally relevant (Elazar et al., 2021; Ravfogel et al., 2021). Accordingly, we do not treat a decodable student attribute as sufficient evidence for a functional student state.

Our architecture goes beyond extracting a token activation from a complete history. It recurrently updates a compact continuous state and removes the raw history from the prediction path. Full-context distillation tests whether this state preserves behaviorally useful information, while future-profile supervision ties the state to later student outcomes. At evaluation, a frozen readout receives no history or student ID; permuting state ownership across held-out students therefore tests whether the student-specific representation matters. Disjoint-history views test within-student stability, and population and same-window outcome baselines test incremental utility. We separately report donor, same-profile, random, and no-op text-transport interventions as diagnostics, without requiring arbitrary cross-student latent transplants to be in distribution.

3 Causally Grounded Generative Student Modeling

3.1 Problem formulation

Let $x_t = (q_t, r_t, y_t, k_t, m_t)$ denote an observed interaction at time t: problem content q_t, student response r_t, correctness y_t, available knowledge-component metadata k_t, and optional interaction metadata m_t. Fields unavailable in a source are omitted rather than imputed. The history before a target is $H_t = (x_1, \ldots, x_{t-1})$. Given a task instruction τ and a target query $\tilde{q}_t$, we predict a structured textual output o_t.

Standard generative prompting estimates

$$p_\theta(o_t \mid \tau, H_t, \tilde{q}_t), \tag{1}$$

which permits the decoder to retrieve arbitrary history tokens and does not identify a student state. We instead factor prediction through a fixed-size continuous state:

$$z_t = U_\theta(z_{t-1}, x_{t-1}), \tag{2}$$

$$p_\theta(o_t \mid \tau, z_t, \tilde{q}_t) = G_\theta(\tau, z_t, \tilde{q}_t). \tag{3}$$

The updater U_θ and decoder G_θ share the same language-model backbone and parameter-efficient adapters. The state $z_t \in \mathbb{R}^{L \times S \times d}$ retains a layerwise trajectory with $S = 8$ slots, all addressed by one repeated <STUDENT_STATE> token type. It is a per-student, time-indexed activation, not a learned student-ID embedding.

The tasks differ only in when the state is read and which output fields are requested. T1 and T2 read the state before the target interaction; T3 first updates the state with the observed wrong response and then requests a misconception; T4 reads the current state but queries a problem Δ interactions in the future; and strict T5 updates with the current wrong response before predicting its recurrence on a later exposure. T6 predicts a future-KC-performance band and is reported only as an appendix proxy, never as a direct latent-mastery label.

3.2 A recurrent state carried by eight slots

We implement the updater with two banks of eight occurrences of the same special token type. For an update, the model receives

$$[\text{<STUDENT_STATE>}_{\text{in}}^{1:S};\ \text{serialize}(x_t);\ \text{<STUDENT_STATE>}_{\text{out}}^{1:S}]. \tag{4}$$

All positions use the same tokenizer ID. At the input positions, we replace the ordinary token embeddings by projected and normalized copies of the previous state,

$$e_{\text{in}}^{1:S} = P_{\text{in}}\,\text{LN}(z_t). \tag{5}$$

The causal hidden activations at the output positions can attend to the previous state and the new interaction; their layerwise trajectory becomes

$$z_{t+1} = P_{\text{out}} h(\text{<STUDENT_STATE>}_{\text{out}}^{1:S}). \tag{6}$$

For the first observed interaction, z_0 is initialized by a learned base state associated with <STUDENT_STATE>. The small projections P_{in} and P_{out}, layer-normalization parameters, and language-model adapters are trained jointly.

At prediction time, the decoder receives

$$[\text{instruction}(\tau);\ \text{<STUDENT_STATE>}^{1:S};\ \text{serialize}(\tilde{q}_t);\ o_t]. \tag{7}$$

The embeddings at <STUDENT_STATE> are replaced by $P_{\text{in}}\,\text{LN}(z_t)$. The raw history is not included, so every history-dependent prediction must pass through the state. This removes the history bypass that weakens one-shot bottleneck masks. It also makes inference incremental: a serving system stores one vector per student, updates it after each interaction, and reuses it for any task query.

We unroll updates over chronological blocks and truncate gradients every B interactions. The state value is preserved across block boundaries while its gradient is detached. This bounds memory independently of the full student history and permits training the 32B backbone with low-rank adapters.

3.3 Multi-task structured generation

All targets use a small structured output language. Binary outcomes use fields such as <correctness>; response tasks use <predicted_response>; misconception diagnosis uses <misconception>; and prospective tasks combine multiple fields. The generative objective is the target-token negative log likelihood

$$\mathcal{L}_{\text{gen}} = -\sum_{j=1}^{|o_t|} \log p_\theta(o_{t,j} \mid o_{t,<j}, \tau, z_t, \tilde{q}_t). \tag{8}$$

Instruction, state, and query tokens are masked from the loss. Task and source samplers are separated so that a large correctness dataset does not dominate open-response or misconception supervision.

This schema unifies the interface, not the construct or metric. Closed-choice responses remain distinct from open mathematical responses; Eedi diagnosis is evaluated against its full taxonomy or by open generation; and strict wrong-response recurrence is not merged with repeat-incorrectness proxies.

3.4 Predictive state distillation

A strict bottleneck can lose information before the updater learns useful compression. We therefore initialize a full-context teacher T from the strong unrestricted multi-task model. For the same target, the teacher sees H_t as in Equation 1, while the student decoder sees only z_t. We minimize a temperature-scaled token-level divergence

$$\mathcal{L}_{\text{KD}} = \mathcal{T}^2 \sum_j \text{KL}\Big(p_T^{(\mathcal{T})}(\cdot \mid H_t, \tau, \tilde{q}_t, o_{t,<j}) \parallel p_\theta^{(\mathcal{T})}(\cdot \mid z_t, \tau, \tilde{q}_t, o_{t,<j})\Big). \tag{9}$$

The teacher is frozen. Ground-truth generation remains the primary objective, so the student is not forced to reproduce teacher errors. The gap between the teacher and state-only student is also reported as an empirical measure of state sufficiency.

3.5 Recurrent–direct state consistency

Errors can accumulate when the state is updated over a long trajectory. We construct a second state from the complete visible history using a direct history encoder E_θ with the existing late-fusion mask. Its state token can attend to all of H_t, but the answer cannot. For selected checkpoints in an unrolled trajectory, we align the recurrent state z_t^{rec} with the stop-gradient direct state $z_t^{\text{dir}} = E_\theta(H_t)$:

$$\mathcal{L}_{\text{dyn}} = 1 - \cos\big(\text{LN}(z_t^{\text{rec}}), \text{sg}[\text{LN}(z_t^{\text{dir}})]\big). \tag{10}$$

We reverse the stop-gradient direction on half of the minibatches. This symmetric schedule prevents one branch from remaining an untrained target while avoiding immediate representational collapse. Both states must also predict the same observed target through Equation 8; cosine agreement alone is never optimized without behavioral supervision.

3.6 Counterfactual grounding with executable misconceptions

Decodability does not ensure that changing a state changes behavior. We use MalruleLib to construct supervised interventions. Let histories H^A and H^B instantiate different misconception rules a and b, and let q be a held-out problem for which both rules are executable. The library provides counterfactual responses $r_a(q)$ and $r_b(q)$. We decode the same query after swapping the two states and optimize

$$\mathcal{L}_{\text{cf}} = -\log p_\theta(r_a(q) \mid z^A, \tau_{\text{response}}, q) \tag{11}$$

$$-\log p_\theta(r_b(q) \mid z^B, \tau_{\text{response}}, q). \tag{12}$$

The pairing procedure deliberately changes the problem template, so the model cannot satisfy the loss by copying a previous answer. We do not attach a malrule classifier to z: the supervision specifies what the state should do, not which label a probe should recover. Evaluation uses unseen history–query pairs, held-out templates, and, where coverage permits, held-out malrules.

3.7 Future-profile ownership grounding

To make student identity testable without exposing a student ID, we add an auxiliary readout over a fixed projection $\phi(z_t)$ of the layerwise State8 trajectory. For student i, let $y_i \in [0,1]^K$ be smoothed correctness rates computed only from post-cutoff interactions and let m_i mark KCs with sufficient support. A linear head predicts

$$\hat{y}_i = \sigma(W\phi(z_i) + b). \tag{13}$$

For a hard mismatched student j, selected inside the training split, we optimize supported future-rate error together with an ownership margin:

$$\mathcal{L}_{\text{prof}} = \text{MSE}_{m_i}(\hat{y}_i, y_i), \tag{14}$$

$$\mathcal{L}_{\text{own}} = \max\{0, \gamma + \text{MSE}_{m_i \cap m_j}(\hat{y}_i, y_i) - \text{MSE}_{m_i \cap m_j}(\hat{y}_i, y_j)\}. \tag{15}$$

Post-cutoff targets supervise training but are never visible to the updater or decoder at inference. KCs below a threshold fixed from training support are excluded from this auxiliary loss.

The complete objective is

$$\mathcal{L} = \mathcal{L}_{\text{gen}} + \lambda_{\text{KD}}\mathcal{L}_{\text{KD}} + \lambda_{\text{dyn}}\mathcal{L}_{\text{dyn}} + \lambda_{\text{cf}}\mathcal{L}_{\text{cf}} + \lambda_{\text{prof}}\mathcal{L}_{\text{prof}} + \lambda_{\text{own}}\mathcal{L}_{\text{own}}. \tag{16}$$

We first warm-start on unrestricted multi-task generation, then train recurrent and direct state paths with distillation and consistency, and finally mix counterfactual and profile batches with real-data replay. This curriculum separates language/output-format learning from state compression and reduces catastrophic forgetting during causal grounding.

4 Experiments

[Draft: Populate from the formal evaluation manifest, not the obsolete six-task matrix. Report every task by source and protocol.]

4.1 Research Questions

RQ1: Generality. Can one model remain competitive across correctness, response, misconception, and prospective-behavior tasks?

RQ2: Capability. Can the model generate concrete responses, natural- language misconception diagnoses with evidence, prospective behavior, and persistence judgments that correctness-only KT does not natively expose?

RQ3: Transfer. Does training on one educational task or supervision type improve a distinct held-out educational task under exposure-matched controls, including but not limited to synthetic-to-real diagnosis?

RQ4: State validity. Does a frozen history-closed State8 predict and explain future behavior for unseen students, does pairing the state with its owner's outcomes matter, and are predictions stable across disjoint views of the same observed history?

4.2 Datasets and Protocols

[Draft: Add the 259,845-example training table and 21,676-example formal evaluation table, then separate T1, T2-choice, T2-open, T3-taxonomy, T3-generation, T4 by prediction distance, and strict T5. Report T6 future-KC-performance only in the appendix.]

4.3 Baselines and Metrics

[Draft: Describe IRT, PFA, BKT, DKT, AKT, conditional response models, full-taxonomy retrieval, and persistence logistic regression. Do not compare LLM-judge semantic accuracy and exact taxonomy accuracy as if identical.]

4.4 Main Results

[Draft: Insert multi-seed results and confidence intervals. Current formal matrix is suitable as a preliminary run, not a final ICLR table.]

4.5 Method Ablations

We compare: (i) the unrestricted full-context model; (ii) the existing one-shot late-fusion bottleneck; (iii) the recurrent architecture with only $\mathcal{L}_{\text{gen}}$; (iv) variants removing distillation, trajectory consistency, or counterfactual grounding; and (v) the complete CGSM. The primary method diagnostic is the state-compression gap between the frozen full-context teacher and the state-only decoder on identical examples. We also report accuracy or generation quality as a function of history length and number of recurrent updates.

4.6 Transfer and Student-State Analyses

[Draft: Add exposure-matched C3 ablations; State8 versus population and same-window baselines; state-ownership permutation tests; disjoint-history stability; readable future profiles; recurrent state drift; and one-shot versus recurrent state comparisons. Report literal donor, same-profile, random, and no-op swaps as diagnostics rather than C4 gates.]

4.7 Controlled Evidence for a One-Shot Causal State

We separately audit the strongest completed one-shot bottleneck, without using it as evidence for recurrent closure. On each frozen intervention pair, the native recipient prediction is the recipient-rule response B. We then replace the residual stream at the single state-token address with either the matched donor trajectory, norm-matched Gaussian residual noise, a same-rule donor trajectory, or the recipient's own trajectory (no-op). The primary contrast is

$$D_{\text{specific}} = \Pr(\widehat{r} = r_A \mid \text{directed donor}) - \Pr(\widehat{r} = r_A \mid \text{Gaussian noise}), \tag{17}$$

computed only on pairs for which the native model predicts B. Here "random" is deliberately an out-of-distribution Gaussian-noise control, not a randomly selected student's state.

5 Conclusion

[Draft: Finalize after the full evaluation matrix and multi-seed C3/C4 experiments are complete.]

We presented a unified generative formulation for student modeling that moves beyond correctness-only prediction toward concrete responses, misconceptions, and prospective behavior. The framework also exposes a designated internal student state that can be evaluated with both probes and causal interventions. For the primary state-validity claim, the intervention is a held-out state-ownership permutation under a closed readout; literal cross-student text transport remains diagnostic. The current evidence supports the feasibility of this approach while underscoring the importance of construct-aware evaluation: open generation is not candidate retrieval, future performance is not a direct mastery label, and decodability is not causality.

References

Ghodai Abdelrahman, Sherif Abdelfattah, Qing Wang, and Yu Lin. DBE-KT22: A knowledge tracing dataset based on online student evaluation. arXiv preprint arXiv:2208.12651, 2022. URL https://arxiv.org/abs/2208.12651.

Xinghe Chen, Naiming Liu, and Shashank Sonkar. MalruleLib: Large-scale executable misconception reasoning with step traces for modeling student thinking in mathematics. arXiv preprint arXiv:2601.03217, 2026. URL https://arxiv.org/abs/2601.03217. Accepted to ACL 2026 Main Conference.

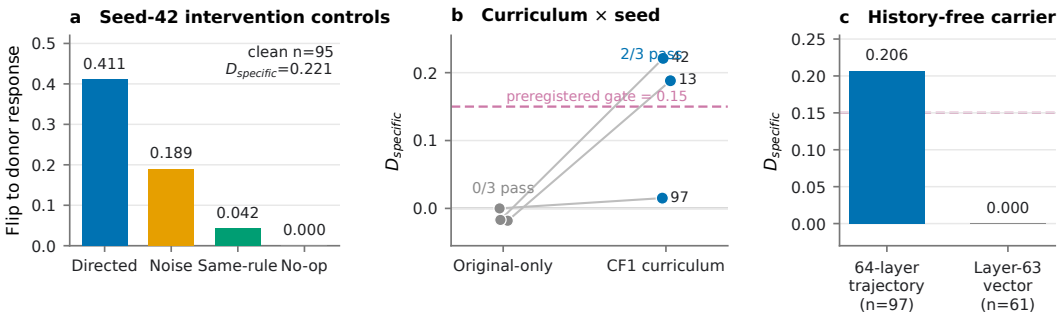


Figure 1: A causally specific one-shot student-state route is conditional on training and is carried across depth. (a) On the canonical CF1 + vision-pad seed-42 checkpoint, replacing the state-token trajectory with a matched donor trajectory changes the recipient prediction to the donor response in 0.411 of the clean pairs, versus 0.189 for norm-matched Gaussian noise, 0.042 for a same-rule donor, and 0.000 for no-op ($n = 95$; $D_{\text{specific}} = 0.221$). (b) The preregistered threshold $D_{\text{specific}} \geq 0.15$ is met by two of three historical CF1 runs but by none of the original-only runs; all seed-level results remain visible. Thus the counterfactual curriculum is necessary in this factorial but is not sufficient across optimization runs. The historical trainer did not seed model and LoRA construction before module initialization, so these are run-level rather than fully controlled training-seed replications. (c) In a strict zero-training bridge, the decoder receives no raw history and has no persistent history KV cache. Transferring the same address across all 64 residual layers preserves causal specificity ($D_{\text{specific}} = 0.206$, $n = 97$), whereas transferring only its final layer-63 vector causes no prediction changes ($D_{\text{specific}} = 0$, $n = 61$). The completed evidence therefore supports a one-address, depth-indexed residual trajectory, not a compact final-layer vector or recurrent/Markov state.

Albert T. Corbett and John R. Anderson. Knowledge tracing: Modeling the acquisition of procedural knowledge. User Modeling and User-Adapted Interaction, 4(4):253–278, 1995. doi: 10.1007/BF01099821.

Jimmy de la Torre. DINA model and parameter estimation: A didactic. Journal of Educational and Behavioral Statistics, 34(1):115–130, 2009. doi: 10.3102/1076998607309474.

Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. Transactions of the Association for Computational Linguistics, 9:160–175, 2021. doi: 10.1162/tacl_a_00359.

Lingyue Fu, Hao Guan, Kounianhua Du, Jianghao Lin, Wei Xia, Weinan Zhang, Ruiming Tang, Yasheng Wang, and Yong Yu. SINKT: A structure-aware inductive knowledge tracing model with large language model. In Proceedings of the 33rd ACM International Conference on Information and Knowledge Management, 2024. doi: 10.1145/3627673.3679760.

Aritra Ghosh, Neil Heffernan, and Andrew S. Lan. Context-aware attentive knowledge tracing. In Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 2330–2339, 2020. doi: 10.1145/3394486.3403282.

John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, pp. 2733–2743, 2019. doi: 10.18653/v1/D19-1275.

Heeseok Jung, Jaesang Yoo, Yohaan Yoon, and Yeonju Jang. CLST: Cold-start mitigation in knowledge tracing by aligning a generative language model as a students' knowledge tracer. arXiv preprint arXiv:2406.10296, 2024. URL https://arxiv.org/abs/2406.10296.

Unggi Lee, Jiyeong Bae, Dohee Kim, Sookbun Lee, Jaekwon Park, Taekyung Ahn, Gunho Lee, Damji Stratton, and Hyeoncheol Kim. Language model can do knowledge tracing:

Simple but effective method to integrate language model and knowledge tracing task. arXiv preprint arXiv:2406.02893, 2024. URL https://arxiv.org/abs/2406.02893.

Zitao Liu, Qiongqiong Liu, Teng Guo, Jiahao Chen, Shuyan Huang, Xiangyu Zhao, Jiliang Tang, Weiqi Luo, and Jian Weng. XES3G5M: A knowledge tracing benchmark dataset with auxiliary information. In Advances in Neural Information Processing Systems Datasets and Benchmarks Track, pp. 32958–32970, 2023. doi: 10.52202/075280-1429.

Shalini Pandey and George Karypis. A self-attentive model for knowledge tracing. In Proceedings of the 12th International Conference on Educational Data Mining, pp. 384–389, 2019. URL https://arxiv.org/abs/1907.06837.

Philip I. Pavlik, Hao Cen, and Kenneth R. Koedinger. Performance factors analysis—a new alternative to knowledge tracing. In Proceedings of the 14th International Conference on Artificial Intelligence in Education, pp. 531–538, 2009. doi: 10.3233/978-1-60750-028-5-531.

Chris Piech, Jonathan Bassen, Jonathan Huang, Surya Ganguli, Mehran Sahami, Leonidas J. Guibas, and Jascha Sohl-Dickstein. Deep knowledge tracing. In Advances in Neural Information Processing Systems, volume 28, 2015. URL https://proceedings.neurips.cc/paper/2015/hash/bac9162b47c56fc8a4d2a519803d51b3-Abstract.html.

Georg Rasch. Probabilistic Models for Some Intelligence and Attainment Tests. Danish Institute for Educational Research, 1960.

Shauli Ravfogel, Grusha Prasad, Tal Linzen, and Yoav Goldberg. Counterfactual interventions reveal the causal effect of relative clause representations on agreement prediction. In Proceedings of the 25th Conference on Computational Natural Language Learning, pp. 194–209, 2021. doi: 10.18653/v1/2021.CoNLL-1.15.

Shashank Sonkar and Richard G. Baraniuk. Deduction under perturbed evidence: Probing student simulation capabilities of large language models. arXiv preprint arXiv:2305.14507, 2023. URL https://arxiv.org/abs/2305.14507.

Jill-Jênn Vie and Hisashi Kashima. Knowledge tracing machines: Factorization machines for knowledge tracing. In Proceedings of the AAAI Conference on Artificial Intelligence, volume 33, pp. 750–757, 2019. doi: 10.1609/aaai.v33i01.3301750.

Zichao Wang, Angus Lamb, Evgeny Saveliev, Pashmina Cameron, Yordan Zaykov, José Miguel Hernández-Lobato, Richard E. Turner, Richard G. Baraniuk, Craig Barton, Simon Peyton Jones, Simon Woodhead, and Cheng Zhang. Results and insights from diagnostic questions: The NeurIPS 2020 education challenge. arXiv preprint arXiv:2104.04034, 2021. URL https://arxiv.org/abs/2104.04034.

Eamon Worden, Cristina Heffernan, Neil Heffernan, and Shashank Sonkar. FoundationalASSIST: An educational dataset for foundational knowledge tracing and pedagogical grounding of large language models. In Proceedings of the International Conference on Artificial Intelligence in Education, 2026. doi: 10.1007/978-3-032-29755-6_36. URL https://arxiv.org/abs/2602.00070.

Tao Wu, Jingyuan Chen, Wang Lin, Mengze Li, Yumeng Zhu, Ang Li, Kun Kuang, and Fei Wu. Embracing imperfection: Simulating students with diverse cognitive levels using LLM-based agents. arXiv preprint arXiv:2505.19997, 2025. URL https://arxiv.org/abs/2505.19997.

Zhihao Yuan, Yunze Xiao, Ming Li, Weihao Xuan, Richard Tong, Mona T. Diab, and Tom M. Mitchell. Towards valid student simulation with large language models. arXiv preprint arXiv:2601.05473, 2026. URL https://arxiv.org/abs/2601.05473.

Jiani Zhang, Xingjian Shi, Irwin King, and Dit-Yan Yeung. Dynamic key-value memory networks for knowledge tracing. In Proceedings of the 26th International Conference on World Wide Web, pp. 765–774, 2017. doi: 10.1145/3038912.3052580.

Table 1: Datasets and task protocols in the unified student-modeling benchmark. N_{tr}, N_{va}, and N_{te} are post-filtering example counts. Real educational logs and controlled synthetic data are reported separately; a dash denotes that a source does not instantiate the task.

Task	Source	Data type	Protocol / target	N_{tr}	N_{va}	N_{te}	Primary metric
T1: correctness	DBE-KT22	real, temporal	next-interaction first-response correctness; new-student/warm-item	3,515	221	225	Accuracy, AUROC
	Eedi Response	real, non-temporal	held-out response conditioned on the same student's profile	15,736	984	984	Accuracy, AUROC
	FoundationalASSIST	real, temporal	identifiable first-response correctness after support/privacy filtering	30,009	1,863	1,882	Accuracy, AUROC
	XES3G5M	real, temporal	next-interaction correctness; official temporal test students	52,009	2,888	7,222	Accuracy, AUROC
	MalRule	synthetic	systematic-rule correctness; same- and cross-template tests	3,000	300	600	Accuracy
T2: response	DBE-KT22	real, temporal	selected-option generation	3,472	218	223	Choice Acc. / MathEq
	Eedi Response	real, non-temporal	profile-held-out option generation	15,736	984	984	Choice Acc. / MathEq
	FoundationalASSIST	real, temporal	privacy-filtered structured mathematical first-response generation	29,995	1,864	1,886	MathEq
	MalRule	synthetic	executable systematic-error response; same- and cross-template tests	3,000	300	600	MathEq
T3: diagnosis	Eedi-Mining	real, response-level	open diagnosis against a 2,587-label expert misconception taxonomy; official fold 4 test	3,476	0	894	Semantic judge; retrieval MRR/$R@k$
	MalRule	synthetic	executable misconception identity; same- and cross-template tests	3,000	300	600	Label Acc.; semantic judge
T4: prospective behavior	DBE-KT22	real, temporal	correctness and selected response on a specified item at $\Delta = 3$	3,186	200	200	JointMath; CorrAcc
	FoundationalASSIST	real, temporal	correctness and open response on a specified item at $\Delta \in \{1, 3, 5\}$	70,811	4,403	4,400	JointMath; CorrAcc
T5: error persistence	DBE-KT22	real, temporal	same question and same wrong option recurs within the next three same-question records	114	15	12	AUROC, AUPRC, Acc.
	FoundationalASSIST	real, temporal	exact normalized wrong first response recurs within the next three same-question records	7,272	974	964	AUROC, AUPRC, Acc.
			Total unique task examples	244,331	15,514	21,676	

Protocol notes. All real longitudinal splits are student-disjoint. Eedi Response lacks timestamps in the public mirror and is therefore a non-temporal profile-prediction protocol, not next-step KT. Eedi-Mining is a response-level diagnosis set and contains no longitudinal student history. MalRule is controlled synthetic evidence used for sanity checks, cross-template generalization, transfer, and mechanism experiments; it is not pooled with real data when making real-world performance claims. T6 (future-KC-performance band; legacy T4) is an appendix-only proxy and is not included in the headline T1–T5 benchmark. Training counts use the original training partition; the final one-epoch fit merged train and validation to 259,845 rows, while the frozen 21,676-row test set was never used for training.

Table 2: Appendix-only T6 future-KC-performance proxy. The target is the three-way band of correctness over up to five subsequent opportunities on the same KC; it is not a direct latent-mastery label and is excluded from the headline T1–T5 totals.

Source	N_{tr}	N_{va}	N_{te}	Primary metrics
DBE-KT22	2,217	139	131	Macro-F1, balanced Acc., QWK
XES3G5M	20,561	1,131	2,808	Macro-F1, balanced Acc., QWK
Total	22,778	1,270	2,939	

A Task Prompts and Output Schemas

[Draft: Add one complete prompt and target example for each protocol.]

Table 3: Source-level scope and split construction. Question and student counts describe the processed source before the final context-length filter.

Source	Tasks used	Students	Questions / labels	History cap	Split and scope
DBE-KT22	T1, T2, T4, T5	1,193	212 questions	6	954/119/120 train/validation/test students; temporal new-student, predominantly warm-item evaluation; multiple-choice database-education interactions.
Eedi Response	T1, T2	4,918	445 questions	6	3,934/492/492 users; user-disjoint deterministic split. The mirror omits timestamps, so held-out examples measure profile-conditioned prediction rather than temporal forecasting.
FoundationalASSIST	T1, T2, T4, T5	5,000	2,986 questions	12	4,000/500/500 students; temporal student-disjoint split. Free-text targets are conservatively filtered to privacy-safe structured mathematics; $\Delta = 1, 3, 5$ for T4.
XES3G5M	T1	18,066	4,137 questions	6	13,008/1,445/3,613 students; official temporal test users. The data record correctness but not the student's selected response, so they do not instantiate headline T2 or strict T5.
Eedi-Mining	T3	–	2,587 labels	–	4,370 question–wrong-answer diagnosis queries; folds 0–3 (3,476) train and official fold 4 (894) test. No student sequence is present.
MalRule	T1, T2, T3	synthetic	21 rules; 132 surfaces	4	3,000/300/600 synthetic profiles; the test combines 300 same-template and 300 cross-template profiles across fractions, decimals, and subtraction.

B Construct Validity and Data Audits

[Draft: Add the remaining construct-validity audit beyond the dataset and task protocol tables.]